André Monteiro

andremgmonteiro01@gmail.com | (+351) [redacted] | Braga, Portugal

mastardy.github.io | github.com/Mastardy

Profile

Software Engineer with nearly 3 years of experience building real-time, interactive 3D applications and visualization tools. Specialised in graphics programming, render optimization, and tooling in C++ and C#, with hands-on experience across Unity and Unreal Engine. Passionate about visualization components, GPU/CPU performance, and understanding systems from the ground up.

Professional Experience

Software Engineer — 10/2025 – Present
EDIGMA — Braga, Portugal

- Developed an interactive 3D application for a museum, featuring an embedded ecosystem-building game and a Qt based CMS enabling museum staff to update content independently
- Built an interactive application using LiDAR point cloud data and projector calibration for real-time spatial interaction
- Designed a full-stack control system (web frontend, backend, embedded middleware) for orchestrating scent diffusers in interactive experiences

Software Engineer — 10/2023 – 09/2025
Shadow Profile, LDA. — Famalicão, Portugal

- Shipped 10+ cross-platform 3D applications in Unity and Unreal Engine across construction (AR/VR), retail, and the games industry
- Profiled and optimized rendering across Unity Projects to hit target framerates on Android, Web (WebGPU) and Windows builds
- Built real-time networked systems using REST APIs, databases, and sockets, powering online experiences for thousands of concurrent users
- Built an in-engine plugin for runtime asset streaming via AWS, reducing app footprint from multiple GB to ~100MB while preserving access to hundreds of models and textures
- Delivered AR/VR visualization tools for the construction industry, working with high-fidelity 3D assets and architectural data

Education

M.Eng. Game Development (Pending Thesis) — 10/2022 – Present
Instituto Politécnico do Cávado e do Ave — Barcelos, Portugal

B.D. Game Design — 09/2019 – 07/2022
Instituto Politécnico de Bragança — Mirandela, Portugal

Skills

Programming Languages
C++, C# (Primary) | Python (Scripting & Tooling) | Typescript (Web)

Graphics & 3D
OpenGL, Compute Shaders | Unity, Unreal Engine

Frameworks & Practices
QT | Git, CI/CD, Agile/Scrum

Projects

Solar System Simulator - Unity DOTS 05/2026 – Present

Real-time solar system simulation built with Unity's Data-Oriented Technology Stack.

- Architected for 100,000+ asteroids in the asteroid belt using cache-friendly ECS layouts and Burst-compiled parallel jobs
- Implementing Keplerian orbital mechanics and a floating-origin system to handle solar-system-scale distances without precision loss

Voxel Ray Tracer 09/2025 – 10/2025

PlonqCraft is a real-time voxel ray tracer built with OpenGL and compute shaders.

- Implemented efficient voxel traversal and chunk-based rendering using compute shaders.
- Developed a physically-inspired lighting system with directional light and hard shadows.
- Optimized rendering pipeline for real-time performance and scalability to larger voxel worlds.

Game Boy Emulator 12/2024 – 02/2025

Cycle-accurate Game Boy emulator written in C++23, built to deepen low-level systems understanding.

- Implemented the full Sharp LR35902 CPU instruction set with accurate flag and timing behaviour
- Built an integrated debugger with instruction-level logging and live state inspection, plus a VRAM viewer for tile and tilemap debugging
- Runs test ROMs and homebrew titles; PPU rendering partially implemented

Personal STL (WSTL) 03/2023 – 09/2023

WSTL is a custom Standard Template Library written by myself using modern C++. It's purpose is to implement better solutions for my Projects.

- Implemented Data Structures, Smart Pointers and Allocators optimized for Games and Simulations.

Custom Game Engine (WCGE) 09/2022 – 11/2023

WCGE is a game engine being written by myself using OpenGL and modern C++

- Implemented a Math Library optimized for Games and abstracted OpenGL into a much easier to use Graphics Library.
- Integrated my personal STL (WSTL), having a hands-on showcase of its capabilities.